WORKED SOLUTION

STEP II 2005, Question 14

Probability and Statistics • Mixture Distributions

STEP II 2005, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q14

The key idea

The density is a weighted mixture of two densities we already know well: a standard normal and a uniform on $[0, \lambda]$. Every integral of f therefore splits into a normal piece and a uniform piece, and we can read off each piece from standard facts (total probability 1, mean, $\mathrm{E}(X^2)$) without doing any hard integration. The variance then comes from $\mathrm{E}(X^2) - \mu^2$.

Solution

The constant k, the mean and the variance

Since ϕ and g are both probability density functions, each integrates to 1 over the real line. So

$$1 = \int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = k\,[1 + \lambda \cdot 1] \quad \Longrightarrow \quad k = \frac{1}{1+\lambda}.$$

For the mean, we use the means of the two components: the standard normal has mean 0, and the uniform density g on $[0, \lambda]$ has mean $\frac{1}{2}\lambda$. Hence

$$\mu = \int_{-\infty}^{\infty} x f(x)\,\mathrm{d}x = k\left[\int x\phi(x)\,\mathrm{d}x + \lambda \int x g(x)\,\mathrm{d}x\right] = \frac{1}{1+\lambda}\left[0 + \lambda \cdot \frac{\lambda}{2}\right] = \frac{\lambda^2}{2(1+\lambda)}.$$

In the same way, the standard normal has $\int x^2\phi(x)\,\mathrm{d}x = 1$ (its variance, since its mean is 0), and

$$\int x^2 g(x)\,\mathrm{d}x = \frac{1}{\lambda}\int_0^{\lambda} x^2\,\mathrm{d}x = \frac{\lambda^2}{3}.$$

So

$$\mathrm{E}(X^2) = \frac{1}{1+\lambda}\left[1 + \lambda \cdot \frac{\lambda^2}{3}\right] = \frac{3+\lambda^3}{3(1+\lambda)}.$$

Ask yourself Why can we split $\mathrm{E}(X^2)$ like this, but not the variance?

Expectations are integrals, and integrals are linear, so $\mathrm{E}(X^2)$ is the same weighted combination of the components' second moments as f is of their densities. The variance is not linear (it involves μ^2), so we must combine the second moments first and only then subtract μ^2. Doing so,

$$\begin{aligned}\sigma^2 = \mathrm{E}(X^2) - \mu^2 &= \frac{3+\lambda^3}{3(1+\lambda)} - \frac{\lambda^4}{4(1+\lambda)^2} \\ &= \frac{4(3+\lambda^3)(1+\lambda) - 3\lambda^4}{12(1+\lambda)^2} \\ &= \frac{12+12\lambda+4\lambda^3+4\lambda^4-3\lambda^4}{12(1+\lambda)^2} = \frac{\lambda^4+4\lambda^3+12\lambda+12}{12(1+\lambda)^2},\end{aligned}$$

which is the required result.

The case $\lambda = 2$

Now $k = \frac{1}{3}$ and $\lambda g(x) = 1$ on $[0,2]$, so

$$f(x) = \begin{cases} \frac{1}{3}\phi(x) & x < 0 \text{ or } x > 2, \\ \frac{1}{3}(\phi(x)+1) & 0 \le x \le 2. \end{cases}$$

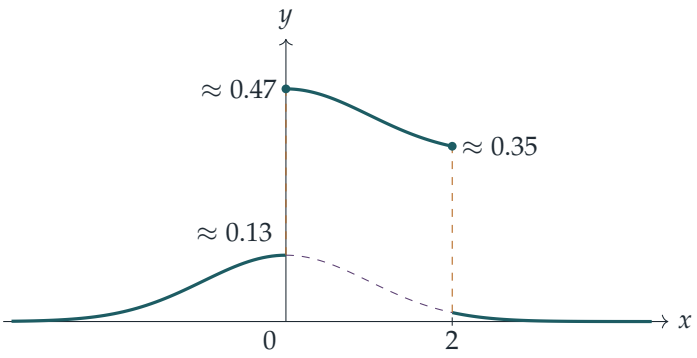
Also

$$\mu = \frac{4}{6} = \frac{2}{3}, \qquad \sigma^2 = \frac{16+32+24+12}{12\cdot 9} = \frac{84}{108} = \frac{7}{9}, \qquad \sigma = \frac{\sqrt{7}}{3}.$$

Part (i)

The curve is a third of the standard normal bell curve, lifted by $\frac{1}{3}$ on the interval $[0,2]$. So it has jumps up at $x = 0$ and down at $x = 2$, each of height $\frac{1}{3}$. Using $\phi(0) = 1/\sqrt{2\pi} \approx 0.399$ and $\phi(2) \approx 0.054$:

- just left of 0, $f \approx 0.133$; just right of 0, $f \approx 0.466$ (the maximum value);
- just left of 2, $f \approx 0.351$; just right of 2, $f \approx 0.018$;
- on $[0,2]$ the curve decreases, following the shape of the normal curve;
- $f \to 0$ as $x \to \pm\infty$, and $f > 0$ everywhere.



The dashed plum curve on $[0,2]$ is $\frac{1}{3}\phi(x)$, shown for comparison; the solid curve is $y = f(x)$. (The vertical scale is stretched.)

Part (ii)

Let $F(x) = \mathrm{P}(X \le x) = \int_{-\infty}^{x} f(t)\,\mathrm{d}t$. Integrating $\frac{1}{3}\phi$ gives $\frac{1}{3}\Phi$, and the extra $\frac{1}{3}$ on $[0,2]$ contributes $\frac{1}{3}$ times the length of $[0,2]$ that lies to the left of x. So

$$F(x) = \begin{cases} \frac{1}{3}\Phi(x) & x < 0, \\ \frac{1}{3}\big(\Phi(x) + x\big) & 0 \le x \le 2, \\ \frac{1}{3}\big(\Phi(x) + 2\big) & x > 2. \end{cases}$$

The pieces agree at $x = 0$ and $x = 2$, as they must (a CDF is continuous when there are no point masses), and $F(x) \to \frac{1}{3}(1+2) = 1$ as $x \to \infty$.

Part (iii)

Here $\mu + 2\sigma = \frac{2}{3} + \frac{2}{3}\sqrt{7} \approx 2.43$, which is greater than 2, so we use the last branch of F there. Since X is continuous,

$$\begin{aligned} \mathrm{P}(0 < X < \mu + 2\sigma) &= F\big(\tfrac{2}{3} + \tfrac{2}{3}\sqrt{7}\big) - F(0) \\ &= \tfrac{1}{3}\big(\Phi(\tfrac{2}{3} + \tfrac{2}{3}\sqrt{7}) + 2\big) - \tfrac{1}{3}\Phi(0) \\ &= \tfrac{1}{3}\big(0.9921 + 2 - 0.5\big) = \frac{2.4921}{3} = 0.8307. \end{aligned}$$

Checks

- With $\lambda \to 0$ the uniform part disappears: the formulae give $\mu \to 0$ and $\sigma^2 \to \frac{12}{12} = 1$, the standard normal values, as they should.
- For $\lambda = 2$, the law of total variance (see below) gives the same answer another way. The components have weights $\frac{1}{3}, \frac{2}{3}$, variances $1, \frac{1}{3}$ and means $0, 1$. The average variance is $\frac{1}{3} + \frac{2}{9} = \frac{5}{9}$, the variance of the means is $\frac{2}{3} - \frac{4}{9} = \frac{2}{9}$, and the total is $\frac{7}{9}$.

- Integrating f numerically (midpoint rule, split at 0 and λ) for $\lambda = 0.5, 1, 2, 3.7$ reproduced total probability 1, the mean and the variance formula to better than 10^{-6}; for $\lambda = 2$ it gave $\mu = 0.6667$ and $\sigma^2 = 0.7778$.
- Computing $\Phi(2.4305)$ directly gives 0.99246 rather than the quoted 0.9921, and the probability 0.8308 rather than 0.8307. The tiny difference comes from the value given in the question, which we should use.

Common slips

- Forgetting the factor $k = \frac{1}{3}$, so that the total probability is 3.
- Computing the variance as $k[\,1 + \lambda \cdot \frac{\lambda^2}{12}\,]$, that is, combining the components' variances. Variances of a mixture do not combine linearly; second moments do.
- Drawing f as a continuous curve: it jumps at $x = 0$ and $x = \lambda$.

Beyond STEP

A density of the form $\sum_i w_i f_i(x)$ with weights $w_i \geq 0$ summing to 1 is a *mixture distribution*: pick component i with probability w_i, then sample from f_i. Here the weights are $\frac{1}{1+\lambda}$ and $\frac{\lambda}{1+\lambda}$. The variance of a mixture is the law of total variance in disguise: the average of the component variances plus the variance of the component means. Mixtures are the basis of clustering methods in statistics and machine learning. Blitzstein and Hwang, *Introduction to Probability*, covers conditional expectation and the law of total variance clearly.

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